Prediction of Anti-Cancer Drug Efficacy Scores

Single-Cell Drug-Response Prediction with Foundation-Model Embeddings

Preliminary Report

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Abstract

This report investigates whether single-cell RNA-seq, combined with foundation-model embeddings, can predict pharmacological response, as a first step toward a reusable pan-cancer single-cell foundation model. On the highest-confidence intersection of the SCP542 single-cell atlas and the CTRPv2 drug-response database (180 cell lines, 545 compounds), we train a shared-trunk multi-task regressor on individual cells and evaluate it under a leave-cell-line-out protocol, comparing scGPT embeddings against a matched principal component analysis (PCA) baseline. The guiding hypothesis is that the foundation model provides a denoised, tissue-agnostic representation of cellular state. With a per-drug-standardized area-under-the-curve target, which is required for the 545-head objective to be learnable, the model ranks unseen cell lines at a Spearman correlation of 0.328 across all drugs and up to 0.396 on a learnable subset, and it exceeds the field-standard NaiveMeanEffects baseline under the DrEval benchmark (normalized Spearman 0.357 for scGPT). Model capacity is not the limiting factor: a ridge regression on cell-line mean embeddings ($\rho = 0.343$) matches the deep model, which indicates that the ceiling is the small number of independent cell lines and the noise of the broadcast bulk labels rather than the representation or the architecture. The scGPT embeddings provide a small advantage over PCA that largely fades under DrEval's normalized metric. We conclude that the approach is viable but currently label-limited, and we outline label- and data-side directions for the next phase. Accordingly, the contribution of this report is not a verdict on whether scGPT outperforms PCA: the available data cannot settle that question, and the margin it does support is small. The contribution is an account of the conditions a study must satisfy before a representation comparison is meaningful at all, and a demonstration that the conventional setup does not satisfy them.

1 Introduction

The overarching goal of this project is to develop a domain-specific, pan-cancer single-cell foundation model capable of predicting pharmacological response, both efficacy and toxicity, that can subsequently be fine-tuned for specific cancer types or for clinical datasets with binary outcomes. Tumor heterogeneity is a major driver of drug resistance and treatment failure, yet this resolution is obscured whenever drug response is modeled from bulk RNA-seq. Predicting response at the single-cell level would allow a model to report a distribution of sensitivities within a sample and, in principle, to identify rare, naturally resistant subclones before selection pressure acts on them. The immediate obstacle is that single-cell drug-screening data does not exist at scale.

This motivates two research questions, and they are the questions the project exists to answer. The first is representational: whether a single-cell foundation model, here scGPT, is a viable basis for drug-effect prediction at all, as opposed to the standard dimensionality reduction that the field defaults to. The second is biological and is the reason the analysis is carried out on individual cells rather than on cell-line averages: whether a model trained on single cells learns cellular heterogeneity *implicitly*, as a by-product of having to map many distinct transcriptional states onto one pharmacological outcome. This second question is the clinically consequential one. Relapse in cancer is typically driven not by the average cell of a tumour but by rare, pre-existing subpopulations that survive treatment and repopulate it; a model that only reproduces the population mean cannot see them, however accurate that mean is. Cell lines are the appropriate testbed for this because they are clonal in origin yet demonstrably heterogeneous in expression [1], so any within-line structure the model exploits is transcriptional state rather than lineage.

To make progress despite this, the project adopts a weakly supervised strategy in which bulk pharmacological labels are mapped onto the individual cells of the corresponding *in vitro* cell lines, and the model is trained to map heterogeneous single-cell inputs to the average bulk response. Laboratory cell lines do not fully recapitulate clinical tumors, but they provide the high-volume, continuous labels required to train a robust representation that can later be fine-tuned on scarcer clinical data. The central methodological difficulty in this setup is a resolution mismatch: standard pharmacological labels, including those of CTRPv2 [2, 3], PRISM [4] and GDSC [5], are derived from bulk cell populations, whereas the transcriptomic data from SCP542 [1] captures individual cellular heterogeneity. Mapping one average bulk score onto hundreds of heterogeneous cells introduces substantial label noise, and regressing raw, sparse single-cell counts directly onto these labels is unstable.

This motivates the use of a foundation-model representation as an intermediate prior. An exploratory latent-space analysis illustrates the effect (Figure 1). When cells are colored by cancer type, full-transcriptome PCA is dominated by tissue-specific markers and separates the data into distinct lineage clusters, so that a downstream model could achieve low error simply by classifying the cell line. The scGPT embeddings [6] instead project the same cells onto a continuous, shared manifold in which tissues are intermixed. The project therefore diverges from prior work such as PERCEPTION [7], which pre-trained on bulk data to predict PRISM response, by focusing on reusable representation learning: it evaluates whether transformer-based single-cell embeddings improve drug-response prediction over a standard PCA baseline, and it is designed to extend to multi-task learning across several databases. The central hypothesis is that scGPT embeddings act as a denoised biological prior that aligns functional cellular states across tissues, so that a regressor trained on this representation is encouraged to learn transcriptomic drivers of sensitivity rather than to memorize the tissue of origin.

This report covers only the initial phase of the work: a single database (CTRPv2), a single continuous response score, single-cell inputs and a supervised regressor. The multi-task run reported here spans 545 drugs but a single database and metric; cross-database integration, toxicity and clinical fine-tuning are left to later work.

A word on what this report can and cannot establish of those two questions. It does not settle the *size* of any scGPT advantage over PCA. The data are too thin for that: the labels are defined per cell line, so the effective sample is roughly 150 independent observations rather than the tens of thousands of cells the model consumes, and the margin the evidence does support is small enough to sit within seed-to-seed variation. Reporting such a margin as a result would be precisely the kind of claim the DrEval study [8] finds the field to be full of. The question this report is organized around is a different and, we argue, prior one: *under what conditions could foundation-model embeddings help in drug-response prediction at all, and why do the conventional evaluation setups fail to test those conditions?* The results below are read in that light. Each central finding is a precondition rather than a performance number: that the multi-task objective must be standardized per drug before any representation can express itself; that a ridge regression on cell-line mean embeddings matches the deep model, so single-cell resolution has not yet earned its place; and that the apparent representation advantage largely dissolves once drug and cell-line mean effects are removed. Establishing these preconditions is the contribution; the comparison itself becomes answerable only once they hold. The second research question, whether heterogeneity is learned implicitly, is likewise not settled here, and for a structural reason set out in the limitations: the present objective penalizes the very within-line variation it would have to

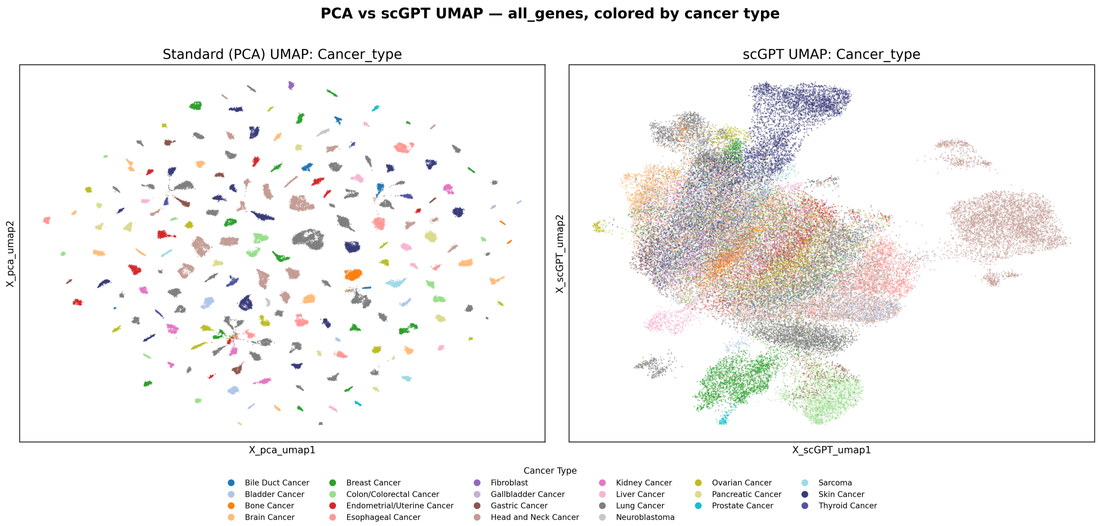

Figure 1: **scGPT removes the tissue-of-origin bias that dominates a PCA baseline.** Cells are embedded by full-transcriptome PCA (left) and by scGPT (right), reduced to two dimensions by UMAP and colored by cancer type. PCA separates the data into tissue-specific clusters, whereas scGPT projects the cells onto a continuous, shared manifold. Produced by `notebooks/06_verify_variants.ipynb`.

express.

2 Methods

+ scGPT erklären wie es funktioniert/trainiert + scGPT wieso? hypothese staten

2.1 Data

All experiments use the highest-confidence intersection of SCP542 [1] and CTRPv2 [2, 3]. This intersection comprises 180 cell lines with post-quality-control measurements (190 roster name matches, of which 10 are unscreened), 545 compounds and 53,513 cells, with effectively complete viability coverage inside the overlap. Genes are reduced to the 5,000 most highly variable; scGPT embeds the 4,576 of these that lie in its vocabulary, while PCA is computed on all 5,000. The gene-set size is not critical: an earlier sweep from 1,000 genes to the full transcriptome showed no advantage for any particular count (`notebooks/07`), so the 5,000-gene selection is used throughout. *Correction, 05.08.2026.* That sweep was computed on the legacy `mean_pv` response score rather than the `auc` target used throughout this report, so its numbers are superseded; it now lives in `notebooks/data_and_harmonization/verify_variants.ipynb`, §9, and the reference to `notebooks/07` above is stale. Its *all-genes* arm is also not the full transcriptome as far as scGPT is concerned: the model reads at most 1,200 genes per cell, drawn at random from that cell's non-zero genes, so what the sweep compared was 5,000 *selected* genes against a random subsample of broadly similar size. The claim that gene-set size does not matter is therefore untested on the current target.

Expression preprocessing (revised 05.08.2026). SCP542 is distributed as counts per million, so library-size normalization has already been applied — once, to the full gene matrix, which is where it belongs. The pipeline therefore does not normalize a second time. Expression is log-transformed as $\log(1 + x)$; the 5,000 most variable genes are selected on that matrix using the dispersion-based Seurat criterion; each gene is then standardized to zero mean and unit variance across cells, with z-scores clipped at 10, before the principal components are taken (scripts/preprocessing/scp542_conversion.py, scripts/preprocessing/add_pca.py).

Until this revision the PCA branch did two things differently. It applied a second library-size normalization *after* gene selection, dividing each cell by the total of its retained genes alone, which rescales a cell by the share of its expression that happens to fall inside the selected set and scales the same cell differently in each gene-set variant. And it applied no per-gene standardization, so the leading components tracked absolute expression level rather than variation between cells. Both are corrected in the code; **every PCA number in this report was produced before the correction** and will change when the representation is recomputed.

scGPT receives the same CPM matrix with neither step applied. It replaces them with a per-cell binning of expression into 51 quantile bins, computed from that cell's own non-zero values — a rank transform, and therefore invariant to any monotone per-cell rescaling. Counts, CPM and $\log(1+x)$ inputs give identical bins, so the CPM input does not disadvantage the model [6].

A property of the data governs every error estimate in this report. The label is defined per cell line and drug, not per cell, so all cells of a line share the same broadcast bulk value and act as pseudo-replicates. The effective sample size is therefore the number of cell lines, approximately 150 after the held-out splits are removed, not the 53,513 cells. Treating cells as independent would understate the uncertainty by roughly the square root of the number of cells per line.

2.2 Representation and model

The comparison between representations is controlled so that only the content of the embedding differs. Both the PCA baseline (`X_pca`) and the scGPT embedding (`X_scGPT`) are 512-dimensional and are fed into the same multi-head perceptron, with hidden dimensions 128 and 64, layer normalization, GELU activations and dropout. The network has one output head per drug and is trained with a masked mean-squared-error loss that is evaluated only on observed cell-drug pairs. The model and training code are defined in `scripts/model/OncoMLP.py` and `scripts/training/train_multitask.py`.

Splits, and where each representation is fitted (added 05.08.2026). Train, validation and test are a 70/15/15 partition of cell lines, so every cell of a line falls on the same side and no line is split across parts. The assignment is now *frozen to a versioned file* (`splits/split_ctrp.csv`) instead of being redrawn from the data on each run. It has to be: eligibility depends on which compounds survive the label filters, so a change to the drug panel would otherwise move lines between train, validation and test unannounced, and results from either side of that change would look comparable while being scored on different held-out lines. Redrawing requires an explicit flag, and a line that the frozen file does not cover is an error rather than a fresh draw.

The principal components used as model input are fitted on the training lines of that split alone — per-gene standardization and the rotation both — and then applied to the held-out lines (`X_pca_train_ctrp`). A separate all-cells decomposition (`X_pca`) is kept for description only, such as the latent-space figures, where holding cells out would be wrong. The cross-validation results are the exception: its folds are drawn at training time rather than stored, so no precomputed train-only projection exists for them and they still use the all-cells decomposition. scGPT needs no such distinction — its weights are pretrained and frozen, and its per-cell binning uses only the cell being embedded, so nothing about it is fitted on this data.

2.3 Response target and per-drug standardization

The choice of training target requires care. CTRPv2 provides a dose-averaged percent-viability score, denoted `mean_pv`, and an integrated area under the fitted dose-response curve, `auc`. The initial experiments used `mean_pv`; the final target is the curve-fit AUC standardized per drug across cell lines, `auc_z`, defined for cell line i and drug j as $\text{auc\_z}_{ij} = (\text{auc}_{ij} - \mu_j)/\sigma_j$, where μ_j and σ_j are the mean and standard deviation of that drug's AUC over the cell lines.

Two considerations motivate the standardization. First, the per-drug response variance is highly heterogeneous across the 545 compounds, with the per-drug standard deviation ranging from 0.034 to 0.302, a factor of roughly 80 in squared error. In a shared multi-task objective with one masked squared-error term per drug, the contribution of a drug scales with its variance, so a small number of high-variance compounds dominate the shared representation irrespective of whether that variance carries any learnable signal; the three drugs with the largest spread (`ifosfamide`, `fqi-2` and `bafilomycin a1`) in fact fail to separate the cell lines at all. Standardizing each drug to unit variance is algebraically equivalent to weighting each head by $1/\sigma_j^2$, which removes this imbalance and is the regression analogue of loss reweighting under class imbalance [9]. Second, per-drug standardization changes the question from absolute potency, which is a property of the drug, to the relative sensitivity of a cell line for a given drug, which is the quantity of interest for personalized prediction. Because the Spearman correlation within a drug is invariant to an affine per-drug transform, the standardization does not alter any single-drug evaluation metric; its only effect is on the multi-task coupling. The target is implemented in `scripts/preprocessing/ctrp_to_h5ad.py`.

2.4 Evaluation

Cell lines are evaluated out of fold. A 5-fold group k-fold split, grouped by cell line, is applied to the 153 training and validation lines, so that no cell of a test line is seen during training. Each line is predicted exactly once, the per-cell predictions of a line are averaged to the line level, and the Spearman correlation between predicted and true values across the roughly 150 lines is computed per drug and averaged over drugs. This protocol replaces an earlier fixed 27-line validation split, on which the standard error of a correlation is approximately 0.2 and effects below 0.4 are not measurable. The Pearson correlation tracks the Spearman value within 0.02 throughout.

For external comparison, the model is additionally evaluated with the DrEval package [8] (version 1.5.1), using its leave-cell-line-out splits, its baselines and its evaluation routine without reimplementation. As recommended by the framework, metrics are reported after the mean drug and cell-line effects have been removed from both the true and predicted responses, and they are pooled over all cell-line-drug pairs rather than averaged per drug, so they are not directly comparable to the per-drug correlations above.

3 Results

Standing of these numbers, 05.08.2026. Every result in this section was produced before the preprocessing corrections recorded in the Methods section, from the representations still on disk. The PCA arm is the one affected: it was computed under a doubly applied library-size normalization and without per-gene standardization. Re-running is deferred until the pipeline review is complete, so that the results are regenerated once from a settled pipeline; see Limitations.

3.1 Per-drug standardization is what makes the multi-task objective learnable

Table 1 compares the three targets across representations and head counts, with all other settings fixed. On the subset of 10 learnable drugs the three targets are statistically indistinguishable, which shows that the curve fit itself buys no accuracy over the dose average. The difference appears only at the full 545-head scale: both unstandardized targets collapse to approximately zero, with scGPT turning negative, whereas the standardized target retains $\rho = 0.328$ (scGPT). Per-drug standardization is therefore a prerequisite for a learnable large multi-task objective rather than a change to the information content of the label.

A set of single-parameter interventions on the unstandardized 545-head configuration locates the cause (Figure 2). Starting from the broken baseline ($\rho = -0.08$), heavier regularization, a

Table 1: Out-of-fold Spearman correlation, averaged over the 10 learnable drugs, seed 42. The three targets tie at K = 10, but only the standardized `auc_z` survives at K = 545. Values are means of the per-drug rows in `outputs/target/target_comparison.csv` (`notebooks/11`).

Heads	Representation	`mean_pv`	`auc`	`auc_z`
10	PCA	0.370	0.360	0.360
10	scGPT	0.396	0.405	0.396
545	PCA	+0.073	−0.012	0.316
545	scGPT	−0.078	−0.069	0.328

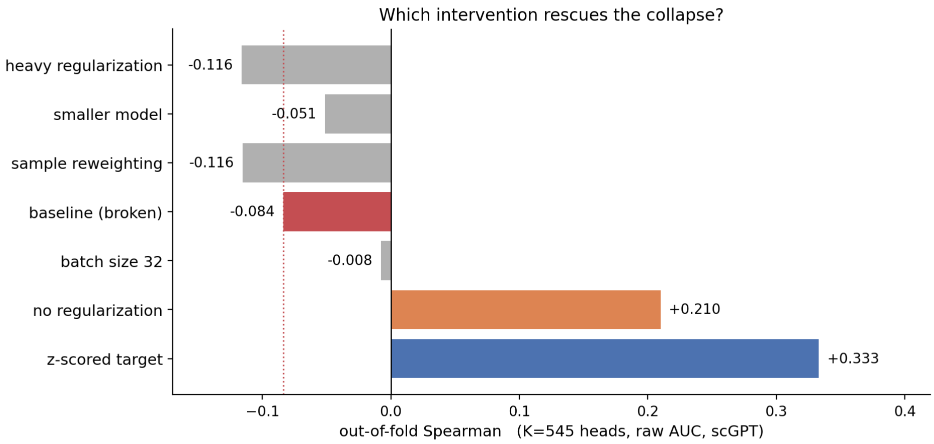

Figure 2: **On the broken 545-head configuration, only removing regularization or standardizing the target recovers the signal.** Each bar applies a single change to the same unstandardized setting (raw `auc`, scGPT). Heavier regularization, a smaller model, a different batch size and reweighting (grey) do not help; removing regularization (orange) recovers most of the gap; per-drug standardization (blue) recovers it fully and addresses the cause. Source: `outputs/ablations/rescue_k545.png` (`notebooks/10`).

smaller model, a different batch size and sample reweighting all leave the result unchanged or make it worse. Two changes help. Removing the regularization recovers most of the gap (to $\rho = +0.21$), which shows that the dropout had been interacting with the mis-scaled loss by starving the learnable heads of shared capacity. Standardizing the target recovers the signal fully (to $\rho = +0.33$) without weakening the regularization, which is why, once the target is corrected, regularization is flat and no longer a useful lever.

3.2 The corrected model ranks unseen cell lines at full head count

With the standardized target the model ranks unseen cell lines well above chance. On the learnable subset it reaches $\rho = 0.396$ (scGPT) and $\rho = 0.360$ (PCA), and it holds at $\rho = 0.328$ and $\rho = 0.316$ even when all 545 heads are trained jointly, which shows that the large multi-task objective is trainable once the target is standardized.

3.3 Model capacity is exhausted; the ceiling is the number of independent cell lines

On the corrected setting, model-side tuning yields nothing further: regularization, capacity (from 74,954 down to 5,130 parameters) and batch size all leave the correlation flat. The decisive control is a ridge regression fitted to the 150 cell-line mean embeddings, with no single cells and no network, which reaches $\rho = 0.343$ (PCA) and essentially ties the PCA multi-layer perceptron

Table 2: DrEval leave-cell-line-out benchmark, normalized metrics, mean over five folds. The model clears the NaiveMeanEffects baseline and edges DrEval's reference models on identical features. Data source: `outputs/dreval/dreval_lco_results.csv` (`notebooks/12`).

Model	norm. Spearman	norm. R^2
NaiveMeanEffects (baseline)	0.000	0.000
SingleDrugRandomForest (scGPT)	0.339	0.098
OncoMLP (X_pca)	0.340	0.086
OncoMLP (X_scGPT)	0.357	0.114

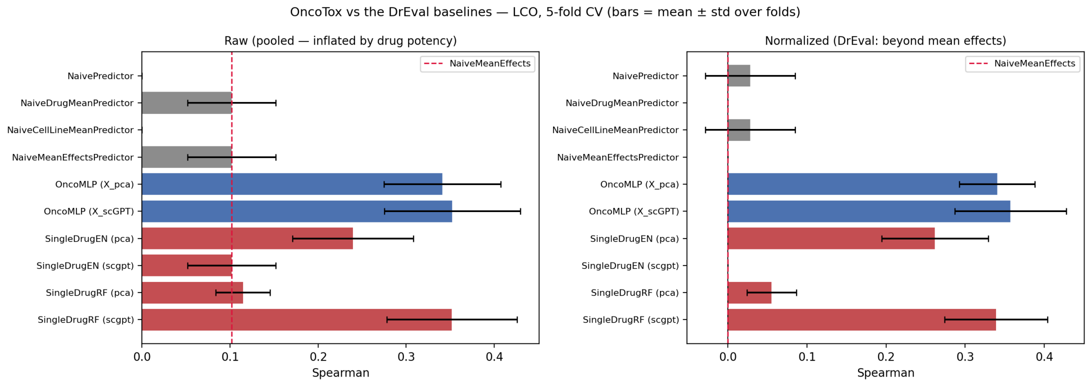

Figure 3: **The naive predictors' raw performance is bias that the normalization removes.** DrEval leave-cell-line-out results, raw pooled correlation (left) and normalized (right). The naive predictors sit above zero on the left and collapse to zero on the right, which shows that their raw performance was the mean effect; the model remains above the baseline after normalization. Source: `outputs/dreval/dreval_lco.png` (`notebooks/12`).

($\rho = 0.356$). The deep single-cell apparatus adds a measurable margin only through scGPT ($\rho = 0.402$), and even there the scGPT signal is nonlinear, since a linear head on scGPT drops to $\rho = 0.292$. The scGPT representation does, however, generalize better than PCA in the way the guiding hypothesis predicts: sharing an identical trunk, its training-to-validation gap on the corrected target is 0.098, against 0.464 for PCA, so the foundation-model embedding overfits far less. The limiting factor is thus the small number of independent cell lines and the noise of the broadcast labels, not the representation or the architecture.

3.4 Against the field, the model clears the naive baseline but not the best-in-class model

Under the DrEval leave-cell-line-out protocol and its normalized metric (Table 2, Figure 3), our model exceeds the NaiveMeanEffects baseline that half of the published field fails to beat, reaching a normalized Spearman correlation of 0.357 for scGPT against 0.000 for the baseline, and it slightly exceeds DrEval's own SingleDrug reference models on identical features ($\rho = 0.339$ for their random forest). It remains below the best model the DrEval study reports, whose normalized coefficient of determination reaches about 19% against our 0.114. The scGPT-over-PCA advantage seen in the internal evaluation shrinks to near noise once the mean effects are removed (normalized Spearman 0.357 versus 0.340). Consistent with the ridge control above, this reproduces the study's central finding that most of the apparent performance in this field is the cell-line and drug mean effect rather than differential response.

3.5 Retiring the standardization: the collapse was a head-count effect

The per-drug standardization was adopted because the untransformed area-under-the-curve target collapsed at full head count. A subsequent run isolates the cause. On an eight-compound panel selected from published sensitivity determinants rather than from our own response statistics, the *untransformed* target reaches a cross-validated per-drug rank correlation of 0.377 for scGPT, against −0.069 for the same target across all 545 heads. The standardization was therefore never repairing the target; it was compensating for pooling compounds whose response variances differ by a factor of 81, and on a panel where that ratio is 2.5 the compensation is unnecessary. Removing the cause is preferable to correcting for it, because dividing by each compound's standard deviation rescales a noise-dominated compound up to full weight in the shared objective, which is a defect of its own.

Two consequences follow for how results are reported. Predictions are now in the units of the assay, so the error is directly interpretable: a mean squared error of 0.0254 against a per-drug-mean null of 0.030 corresponds to roughly 15 % of the variance explained and a typical deviation of 0.16 viability. And the target is the quantity the source database publishes, which makes comparison against DrEval, GDSC and PRISM direct rather than mediated by a project-internal transform.

Table 3: **Out-of-fold performance on the literature panel, untransformed target.** Mean over the eight compounds, five-fold cross-validation grouped by cell line, one seed. Source: `outputs/panel/panel_leaderboard.csv` (`notebooks/14`).

	Spearman		MSE	
Model	PCA	scGPT	PCA	scGPT
MLP, unweighted	0.316 ± 0.003	**0.377**	0.0265	0.0254
MLP, density-weighted	0.308	0.369	0.0274	0.0254
Ridge on cell-line means	0.306	0.299	0.0270	0.0268

The table reports pooled out-of-fold correlations, which the cross-validated design should be qualified with. Recomputing the score within each fold's held-out lines gives a fold-to-fold standard deviation of 0.028 for PCA and 0.043 for scGPT, and the spread across the eight compounds is larger still (0.111 and 0.091, with scGPT ranging from 0.296 to 0.546). The two dispersions answer different questions — how much the result depends on which lines were held out, against how unevenly the model performs across compounds — and neither substitutes for the other. The consequence for the comparison is that the scGPT advantage of 0.062 is of the same order as a single fold standard deviation, which places it as consistent evidence rather than an established margin.

The ridge control behaves as on every previous panel: it ties the PCA model (0.306 against 0.316), while the scGPT model clears it by 0.077. That margin was 0.082 on the earlier, larger panel, so this is a replication rather than a new finding — and a stronger one, because the compounds here were chosen without reference to our labels.

Two properties of the runs qualify how precisely these numbers should be read. The PCA arm is not bit-reproducible: four executions of identical code and seed gave 0.313, 0.315, 0.317 and 0.320, while every other arm reproduced exactly. The cause is visible in the training logs — the PCA model reaches its best validation epoch after a single epoch in four of five folds, against ten to twenty for scGPT — so its selected checkpoint is chosen among near-tied, barely-trained states. That is a mechanistic restatement of the ridge tie rather than a separate observation: a model that peaks after one pass over the data is doing little more than the null predictor it was initialized at. Absolute correlations are lower than on the earlier panel (0.377 against 0.402), which is the expected direction: those compounds had been selected using all cell lines, validation and test included.

3.6 An imbalance correction that did not work

Because most cell lines of a given compound sit in a narrow band of the response range while the pharmacologically interesting extremes are sparse, an unweighted squared-error objective is dominated by the crowded centre. Weighting each observation by the inverse of a smoothed estimate of the label density is the established remedy [10, 11], and it was applied per compound, estimated within each fold on training lines only. It does not help: mean rank correlation moves by -0.006 for PCA and -0.008 for scGPT.

The result is informative rather than merely negative, because the mechanism demonstrably operated. The standard deviation of the predictions rises from 0.062 to 0.082 (PCA) and 0.080 (scGPT), which is the reduced shrinkage the weighting is designed to produce; the objective was reweighted as intended and the ranking simply did not follow. The explanation is visible in the label distributions. Before correction, several compounds appeared strongly right-skewed, but that tail consisted almost entirely of measurements above an area-under-the-curve of 1.0, where the compound apparently improved growth relative to control — assay artifact rather than biology. Once those 1.1 % of measurements are winsorized, every compound has an absolute skewness below 0.5, and an imbalance correction has little imbalance left to act on.

4 Discussion

The results reframe an earlier, apparently negative finding. A first pass of this work had reported that neither representation could rank cell lines, with out-of-fold correlations near zero over the full drug panel. That outcome was substantially an artifact of the training target rather than a statement about the data or the representation. The initial score, a dose-averaged viability, left the per-drug variances untouched, so in the shared multi-task loss a handful of high-variance and largely uninformative compounds dominated the representation. Standardizing the target per drug removes this imbalance, and with it the model ranks unseen cell lines at correlations of 0.328 across the panel and up to 0.396 on a learnable subset. The standardization is best understood as a necessary preprocessing choice for a heterogeneous multi-task objective, not as a scientific result in itself; a head-to-head comparison of the candidate scores is informative mainly because it shows that the curve fit alone changes nothing and the standardization changes everything. The dominance of the target change is direct: on the unstandardized targets the same model collapses at full head count (Table 1), and no other single intervention on the broken configuration approaches it (Figure 2).

The corrected picture is modest but genuine, and it is best read against the standards the DrEval study sets for the field. The model clears the NaiveMeanEffects baseline that about half of published models fail to beat, and it edges DrEval's own reference models on identical features, but it remains below the best model that study reports and shows only a faint representation advantage once the mean effects are normalized away. The most informative internal result points the same way: a ridge regression on cell-line mean embeddings matches the deep model, so the information the current pipeline exploits is largely captured by a per-line average. This is the cell-line-effect bias that DrEval was designed to expose, reproduced here independently. The single-cell resolution and the scGPT embedding together add a small margin over that average, and scGPT needs a nonlinear head to realize it, which is the first concrete evidence that the embedding carries structure a linear model cannot reach. This lower reliance on memorization is the effect the guiding hypothesis anticipated, and it persists on the corrected target, where the scGPT model's training-to-validation gap (0.098) is far smaller than that of PCA (0.464) under an identical trunk.

Taken together, the evidence indicates that the limiting factor at this stage is the label side rather than the model side. The task is defined on roughly 150 independent cell lines, each carrying one bulk value broadcast onto hundreds of cells, and model-side tuning is exhausted across regularization, capacity and batch size. Improving performance therefore requires acting on the labels and the data, not on the architecture.

It is worth stating explicitly what this reframing implies for the question the project set out to ask. The comparison that motivated the work, whether foundation-model embeddings beat a PCA baseline, is not answered here, and on this data it is not answerable: the margin is of the same order as the seed-to-seed variation, it does not survive normalization, and the effective sample size is the number of cell lines rather than the number of cells. What the work does answer is the prior question of what must be true before that comparison carries any information. Three conditions emerge from the results above, and none of them held at the outset. The training objective must be standardized across a heterogeneous set of drug heads, or a minority of high-variance and largely uninformative compounds captures the shared representation and drives every correlation to zero. The evaluation must be scored against a control that already absorbs the cell-line effect, since a ridge regression on cell-line mean embeddings reproduces the deep model's performance and a per-drug-mean null is too weak a bar to distinguish the two. And the reported metric must have the mean effects removed, because a substantial share of the raw correlation is attributable to which drug is potent and which line is broadly sensitive rather than to their interaction. Each condition is a way in which a conventional setup can report an apparent representation advantage that is not one, which is the mechanism the DrEval study [8] identifies across the published literature and which is reproduced here independently on our own data. Read this way, the negative and diagnostic results are not preliminary steps toward the comparison; they are the finding, and the comparison is deferred until the data can support it.

5 Limitations and Future Work

+ add outlook + multi input (GDSC, PRISM REPURPOSED) + MIL/Attention Pooling + scDEAL bulk pretraining + more focus on explainability/cell heterogeneity

Several limitations bound the strength of the present claims, and each points to a concrete next step.

The most serious is that the learnable-drug subset was selected using all 180 cell lines, including the validation and test lines, so any number reported on that subset is a best-case diagnostic rather than a generalization estimate; the selection should instead be performed inside each cross-validation fold, using only the training lines. The subset is moreover chosen from label statistics alone and was not validated in this report against the per-drug performance the model actually achieves, so it should be read as a diagnostic device rather than a claim that only ten drugs are learnable.

The scGPT advantage over PCA is small: a Spearman gap of $+0.036$ at $K = 10$ and $+0.012$ at $K = 545$, of the same order as the seed-to-seed variation. Across three seeds the gap averages $+0.04$ and is not sign-consistent, ranging from -0.00 to $+0.09$, and it does not survive DrEval's normalized metric, so it should not be presented as an established margin until it has been reproduced over more seeds and a wider drug set. The external benchmark is itself preliminary, covering only the leave-cell-line-out split on the learnable subset rather than the leave-tissue-out and leave-drug-out settings or the full panel.

Reproducibility, and the standing of the numbers below (05.08.2026). A review of the preprocessing pipeline found several sources of run-to-run variation that had not been controlled. They are recorded here rather than beside any one result because they qualify all of them. Random seeds were set in the training code but not everywhere randomness was actually consumed: the principal-component decomposition silently used the library default rather than the project seed, the training data loaders took their shuffling order from the global generator instead of an explicit one, and the embedding script set no seed at all. All are now fixed to a single project-wide value.

This matters most for scGPT, whose embedding path is not deterministic by construction. The model reads at most 1,200 genes per cell and samples them at random when a cell has more, so the all-genes variant is a random subsample of each cell's transcriptome rather than the transcriptome itself. With the seed fixed it is one reproducible draw, and it is treated here as a sanity check rather than as a full-transcriptome comparator — a single well-defined input is preferable to averaging over draws, which would multiply the cost of every experiment and leave each individual result harder to attribute.

The hardware also bounds what can be checked. FlashAttention is CUDA-only and cannot be installed on this machine, so scGPT runs through the standard attention implementation; this is numerically equivalent but is why embedding one gene-set variant takes about twenty minutes here, and why the variants are not regenerated on demand. Training on the MPS backend is not bit-reproducible: four identical runs of the PCA arm returned Spearman correlations of 0.313, 0.315, 0.317 and 0.320 while every other arm was exact, so differences smaller than that band are not interpretable.

A second asymmetry concerns how each representation is built, and it bears on the comparison itself. The principal components were fitted on the full expression matrix: gene selection, per-gene standardization and the rotation all used every cell, held-out lines included. The scGPT embedding is not fitted on this data at all — the weights are pretrained and frozen, and the binning that precedes them uses only the cell being embedded — so the two arms were not on equal terms. None of these fits sees a response label, which is why standard pipelines accept them, but the PCA baseline was tuned to the cells it was evaluated on and the scGPT representation was not. The bias runs toward the control, so any scGPT advantage measured this way is conservative on that account.

For the fixed train/validation/test split this is now corrected: standardization and rotation are fitted on the training lines alone and applied to the held-out ones, as described in the Methods. It is *not* corrected for the cross-validated results, whose folds are drawn at training time and so have no stored train-only projection; those remain on the all-cells decomposition, and gene selection remains an all-cells step for both arms. The numbers reported below all predate the correction in any case.

Finally, the preprocessing corrections described in the Methods section change what the code produces, and nothing has been recomputed. **Every number in this report was produced by the pipeline as it stood before those corrections**, from the representations still on disk. They are reported as the record of what was actually run; re-running is deliberately deferred until the pipeline review is complete, so that results are regenerated once from a settled pipeline rather than repeatedly from a moving one. The embedding script was itself, until this review, an untracked file outside the repository, and is now versioned alongside the results it produces.

A further limitation concerns the loss rather than the labels. The response values of a given compound are not uniformly distributed over the cell lines: most lines cluster in a narrow band and the pharmacologically interesting extremes are sparse. Under an unweighted squared-error objective the dense region dominates the gradient, and the optimal predictor shrinks toward each compound's mean, which is exactly the behaviour observed here (predicted standard deviations of 0.53 and 0.47 against a true spread of 1.0). This is the regression analogue of class imbalance, and the established remedy is to weight each observation by the inverse of a smoothed estimate of the label density, so that rare response values contribute in proportion to their scarcity rather than their frequency [10, 11]. This was implemented and tested, and it does not help; the result is reported above. The shrinkage is therefore not an artifact of how the objective weights the response range, which leaves the small number of independent cell lines as the remaining explanation and makes the case for acting on the label side rather than on the loss.

The clearest opportunities for improvement follow from the diagnosis that the task is label-limited. First, the training target can be aligned with the evaluation by regressing on the doubly normalized residual, in which both the drug and the cell-line mean effects are removed, with the means estimated only on the training lines of each fold; this matches the quantity DrEval rewards, and an initial check (`outputs/dreval/dreval_normalized.csv`) indicates that about a fifth of the current signal is the cell-line effect alone. Second, the single-cell resolution has yet to justify itself, since averaging a line's cells loses nothing measurable; a pooling model that predicts the line label from a bag of cells, for example through attention, is the natural test and must beat the ridge control to be worthwhile. This point deserves to be stated more sharply, because it bears directly on the second research question. The pharmacological label is constant within a cell line, and the masked objective therefore requires every cell of a line to regress onto the same scalar. Any difference the model predicts between two cells of the same line is scored as error, so the loss actively penalizes the within-line variation that the single-cell framing exists to capture; under this objective the optimal per-cell predictor is constant across a line, and heterogeneity can only appear as failure to fit. It follows that the present setup does not test whether heterogeneity is learned implicitly — it is structurally unable to. Multiple-instance learning is not merely an incremental architecture to try but the minimal change that makes the question askable: predicting the line label from a bag of cells constrains only the aggregate, leaves the model free to assign different values and weights to different cells, and exposes the attention weights as a readout of which subpopulation drives the response. That readout is the quantity of clinical interest, since relapse is driven by rare surviving subpopulations rather than by the population mean, and it is the natural point at which to connect the model to the recurrent heterogeneity programs annotated for this dataset [1]. Note also that the per-cell loss currently weights each cell line by how many cells were sequenced from it, which ranges from 56 to 1,990 and is an artifact of the assay rather than information about the label; weighting each line equally is a prerequisite for any of these comparisons to be read as biology. Third, and most directly, the ceiling can be raised by adding independent cell lines, as CTRPv2 alone screens roughly 1,100 against the 180 in the current overlap. Beyond these, a diagnostic error analysis of where the model fails is low-risk and immediately useful, whereas gene-level interpretation to surface transcriptomic drivers of resistance is worthwhile only once per-drug performance is substantially higher and stable, so that the interpretation does not merely describe noise. The longer horizon remains a cross-database, multi-task model spanning efficacy and toxicity, fine-tunable on clinical outcomes.

The order of the next two steps is deliberate rather than a matter of convenience. **Multiple-instance learning** comes first because it is the only untested capacity lever and, more importantly, the minimal change that makes the second research question answerable at all: under the present objective every cell of a line is regressed onto the same scalar, so within-line variation is scored as error rather than as the quantity of interest. Predicting a line's response from a bag of its cells constrains only the aggregate, and the resulting attention weights identify which subpopulation drives the response, which is the readout the clinical motivation calls for. It requires no additional data, and it is falsifiable: it must exceed both the per-cell model and the ridge control, and failing both would establish that single-cell resolution does not pay for itself here.

Bulk pretraining in the manner of scDEAL follows. Every remaining lever is on the label side: model-side tuning is exhausted, a ridge regression on cell-line means matches the deep model, and the density weighting reported above was a null. What binds is the small number of independent cell lines, each carrying a single broadcast value. The screens themselves are far larger — CTRPv2 covers roughly 1,100 lines against the 180 in the present intersection — so the labels exist and it is the single-cell expression that is missing. Pretraining a denoising autoencoder on bulk expression and aligning the bulk and single-cell latent spaces by domain adaptation addresses that gap directly, rather than accepting the noise introduced by copying one bulk value onto several hundred cells. It is second because it changes where the representation originates while multiple-instance learning changes how cells map to a line-level prediction; introducing both at once would leave the outcome unattributable, which is the error that made the earlier target defect so costly to diagnose.

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